Vectors and Embeddings in AI

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Vectors & Embeddings in AI Practice Exercises**

Two document embeddings $\mathbf{a} = [5, 10]^T$ and $\mathbf{b} = [10, 5]^T$ are compared. If $||\mathbf{a}|| = ||\mathbf{b}|| = \sqrt{125}$, calculate their exact cosine similarity.

Dot product $\mathbf{a} \cdot \mathbf{b} = (5)(10) + (10)(5) = 100$. Product of norms $||\mathbf{a}||||\mathbf{b}|| = 125$. Cosine similarity = $\frac{100}{125} = 0.800$. Final Answer: 0.800

2 **[Foundational] • Target: Vector Spaces and Linear Combinations**

Consider a dataset of n data points, where each data point is a vector in $\mathbb{R}^D$. A non-linear dimensionality reduction technique maps these data points to a lower-dimensional space $\mathbb{R}^d$. The mapping function for a single data point $\mathbf{x} \in \mathbb{R}^D$ is given by $f(\mathbf{x}) = \phi(\mathbf{x})$, where $\phi : \mathbb{R}^D \to \mathbb{R}^d$ is a function designed to preserve local neighborhood structures. Suppose the mapping ϕ is defined by an explicit feature map $\phi(\mathbf{x}) = [\mathbf{x}^T\mathbf{x}, \mathbf{x}^T\mathbf{x}, \ldots, \mathbf{x}^T\mathbf{x}]^T$ where the vector $\mathbf{x} \in \mathbb{R}^{10}$ and d is the dimension of the embedded space. If the goal is to embed these points into a space such that pairwise Euclidean distances are preserved as closely as possible under a specific distortion, and it is found that $d = 15$ yields optimal results for this mapping under a certain constraint, determine the effective rank of the feature map's image space.

The feature map is defined as $\phi(\mathbf{x}) = [\mathbf{x}^T\mathbf{x}, \mathbf{x}^T\mathbf{x}, \ldots, \mathbf{x}^T\mathbf{x}]^T$, where $\mathbf{x} \in \mathbb{R}^{10}$. The term $\mathbf{x}^T\mathbf{x}$ is the squared Euclidean norm of $\mathbf{x}$, denoted as $||\mathbf{x}||^2$. This is a scalar value. Thus, the mapping ϕ produces a vector where each component is the same scalar value, $||\mathbf{x}||^2$. Let $k = ||\mathbf{x}||^2$. Then $\phi(\mathbf{x}) = [k, k, \ldots, k]^T$ in $\mathbb{R}^d$. Th.

3 **[Foundational] • Target: Vector Norms and Sparsity**

Consider a linear transformation represented by the matrix $M = [2, k; 1, 3]$. A vector $\mathbf{v} = [1; 2]$ is transformed by M into $\mathbf{v}' = M\mathbf{v} = [8; 7]$. Determine the value of the unknown parameter k.

The problem involves a matrix-vector multiplication. We are given the matrix $M = [2, k; 1, 3]$ and the vectors $\mathbf{v} = [1; 2]$ and $\mathbf{v}' = [8; 7]$. The transformation is defined by $\mathbf{v}' = M\mathbf{v}$. Let's perform the matrix-vector multiplication: $[8; 7] = [2, k; 1, 3][1; 2]$ This expands to two equations: 1. $8 = (2)(1) + (k)(2)$ 2. $7 = (1)(1) + (3)(2)$ Let's solve the first equation for k: $8 = 2 + 2k$ Subtract 2.

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4 [Foundational] • Target: Dot Products, Cosine Similarity and Angular Distance

Consider a dataset of observations $\{x_i\}_{i=1}^{n}$ where each x_i is a 2-dimensional vector. The sample covariance matrix for this dataset is given by: $S = (4, -2; -2, 1)$ Principal Component Analysis (PCA) is applied to this dataset. The first principal component corresponds to the direction with the largest variance. Let the direction vector of the first principal component be $\mathbf{v}_1 = (a; b)$. The variance explained by this principal component is given by the corresponding eigenvalue λ_1. If $\mathbf{v}_1$ is a unit vector, what is the value of $a^2 + b^2$?

The first principal component direction vector $\mathbf{v}_1$ is an eigenvector of the covariance matrix S. The variance explained by this component is the corresponding eigenvalue λ_1. The problem states that $\mathbf{v}_1 = (a; b)$ is a unit vector. By definition, a unit vector has a magnitude of 1. The squared magnitude of a vector $(a; b)$ is $a^2 + b^2$. Therefore, if $\mathbf{v}_1$ is a unit vector, its squared magnitude must be 1, which means $a^2 + b^2 = 1$. The specific va.

5 [Foundational] • Target: High-Dimensional Geometry and Projections

A linear transformation is defined by the matrix $M = (2, 1; 3, 4)$. If an input vector is $v = (x; 1)$, and the output vector after the transformation is $w = (5; 7)$, what is the value of x?

The linear transformation is given by $w = Mv$. Substituting the given matrices and vectors, we have: $(3; 7) = (2, 1; 3, 4)(x; 1)$ Performing the matrix multiplication on the right side gives: $(3; 7) = (2x + 1(1); 3x + 4(1)) = (2x + 1; 3x + 4)$ Equating the corresponding components of the vectors, we get two equations: 1. $3 = 2x + 1$ 2. $7 = 3x + 4$ Solving the first equation for x: $3 - 1 = 2x$ $2 = 2x$ $x = \frac{2}{2}$ $x = 1$ Solving.

6 [Foundational] • Target: Vectors & Embeddings in AI Practice Exercises

Consider a linear layer with a weight matrix W and an input vector $\mathbf{x}$. If $W = (1, 2; 3, 4)$ and $\mathbf{x} = (5; 6)$, compute the output vector $\mathbf{y} = W\mathbf{x}$.

The output vector $\mathbf{y}$ is computed by multiplying the weight matrix W by the input vector $\mathbf{x}$. This is a standard matrix-vector multiplication. \\ \\ $\mathbf{y} = W\mathbf{x} = (1, 2; 3, 4)(5; 6)$\\\ \\ To compute the first element of $\mathbf{y}$, we take the dot product of the first row of W with $\mathbf{x}$: $(1)(5) + (2)(6) = 5 + 12 = 17$.\\ \\ To compute the second element of $\mathbf{y}$, we take the dot product of the second row of W with $\mathbf{x}$.

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7 [Foundational] • Target: Vector Spaces and Linear Combinations

A vector $\mathbf{v}$ is transformed into a new vector $\mathbf{v}'$ by a linear transformation represented by the matrix $\mathbf{M}$. We are given that $\mathbf{v} = (2\ 3)$ and $\mathbf{v}' = (7\ 11)$. The transformation matrix is of the form $\mathbf{M} = (a, 1\ 4, b)$. Determine the value of a and b.

The transformation is defined by $\mathbf{v}' = \mathbf{M}\mathbf{v}$. Substituting the given vectors and matrix form: $(7\ 11) = (a, 1\ 4, b)(2\ 3)$ Performing the matrix multiplication: $(7\ 11) = (2a + 3\ 8 + 3b)$ Equating the corresponding components, we get a system of two linear equations: 1) $2a + 3 = 7$ 2) $8 + 3b = 11$ Solving the first equation for a: $2a = 7 - 3$ $2a = 4$ $a = 2$ Solving the second equation for b: $3b = 11 - 8$ $3b = 3$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Vector Norms and Sparsity

Consider two vectors, $\mathbf{u} = [3; 4]$ and $\mathbf{v} = [1; -2]$. What is the vector representing $2\mathbf{u} - \mathbf{v}$?

To find the vector $2\mathbf{u} - \mathbf{v}$, we first compute $2\mathbf{u}$ and then subtract $\mathbf{v}$ from the result. $2\mathbf{u} = 2 \times [3; 4] = [2 \times 3; 2 \times 4] = [6; 8]$. Now, we subtract $\mathbf{v}$ from $2\mathbf{u}$: $2\mathbf{u} - \mathbf{v} = [6; 8] - [1; -2] = [6 - 1; 8 - (-2)] = [5; 8 + 2] = [5; 10]$. Therefore, the resulting vector is $[5; 10]$.

9 [Intermediate] • Target: Dot Products, Cosine Similarity and Angular Distance

Consider two vectors, $\mathbf{u} = (3; -2)$ and $\mathbf{v} = (-1; 5)$. What is the resultant vector $\mathbf{w} = \mathbf{u} + \mathbf{v}$?

To find the resultant vector $\mathbf{w} = \mathbf{u} + \mathbf{v}$, we add the corresponding components of the two vectors. $\mathbf{w} = (3; -2) + (-1; 5) = (3 + (-1); -2 + 5) = (2; 3)$.

10 [Intermediate] • Target: High-Dimensional Geometry and Projections

A simple linear regression model has been fitted to a dataset. The model equation is given by $\hat{y} = ax + b$. The regression line passes through the points $(2, 10)$ and $(5, 22)$ in the feature-target space. Determine the value of the intercept b.

The problem states that the linear regression line $\hat{y} = ax + b$ passes through two points: $(x_1, y_1) = (2, 10)$ and $(x_2, y_2) = (5, 22)$. We can use these two points to form a system of linear equations to solve for the unknown parameters a (the slope) and b (the intercept). Substituting the first point $(2, 10)$ into the equation $\hat{y} = ax + b$ gives: $10 = a(2) + b$ (Equation 1) Substituting the second point $(5, 22)$ into the equation $\hat{y} = ax + b$ gives: $22 = a(5) + b$ (Equation 2) To fi.

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11 **[Intermediate]** • **Target: Vectors & Embeddings in AI Practice Exercises**

A 2D linear transformation T is defined by the matrix $M = (3, 1; 1, 3)$. Consider a triangle with vertices $A = (1, 0)$, $B = (0, 1)$, and $C = (1, 1)$. This triangle is transformed by T to a new triangle with vertices A', B', and C'. The area of the transformed triangle $A'B'C'$ is then passed through a ReLU activation function, $f(x) = \max(0, x)$. What is the final output value?

The area of the original triangle with vertices $A = (1, 0)$, $B = (0, 1)$, and $C = (1, 1)$ can be calculated. We can use the determinant formula for the area of a triangle given its vertices (x_1, y_1), (x_2, y_2), (x_3, y_3): Area = $\frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$. Plugging in the coordinates: Area = $\frac{1}{2}|1(1 - 1) + 0(1 - 0) + 1(0 - 1)| = \frac{1}{2}|0 + 0 - 1| = \frac{1}{2}$. A linear transformation represented by a matrix M scales the area of a 2D shape by the abs.

12 **[Intermediate]** • **Target: Vector Spaces and Linear Combinations**

Consider a 2D dataset represented by the points $A = (1, 2)$, $B = (4, 2)$, and $C = (2, 5)$. If this dataset undergoes a linear transformation represented by the matrix $M = (2, 0; 0, 1)$, what are the coordinates of the transformed points A', B', and C'?

The transformation of a point $\mathbf{x}$ by a matrix M is given by $M\mathbf{x}$. We apply the matrix $M = (2, 0; 0, 1)$ to each point: For point $A = (1, 2)$, the transformed point A' is: $(2, 0; 0, 1)(1; 2) = (2 \times 1 + 0 \times 2; 0 \times 1 + 1 \times 2) = (2; 2)$ So, $A' = (2, 2)$. For point $B = (4, 2)$, the transformed point B' is: $(2, 0; 0, 1)(4; 2) = (2; 2)$ So, $B' = (8, 2)$. For point.

13 **[Intermediate]** • **Target: Vector Norms and Sparsity**

A binary classification model uses a logistic sigmoid function $\sigma(z) = \frac{1}{1 + e^{-z}}$. The decision boundary is defined by $\sigma(z) = 0.5$, where $z = w_1x_1 + w_0$. For a dataset with two features x_1 and x_2, the model's linear component is $z = 2x_1 - x_2 + b$. If the classification boundary passes through the point $(x_1, x_2) = (3, 1)$, what is the value of the bias b?

The logistic sigmoid function is $\sigma(z) = \frac{1}{1 + e^{-z}}$. The decision boundary is where $\sigma(z) = 0.5$. Setting $\sigma(z) = 0.5$ gives $\frac{1}{1 + e^{-z}} = 0.5$, which implies $1 = 0.5(1 + e^{-z})$, so $2 = 1 + e^{-z}$, and $e^{-z} = 1$. This occurs when $-z = 0$, or $z = 0$. Therefore, the decision boundary is defined by $z = 0$. The linear component of the model is given by $z = w_1x_1 + w_2x_2 + b$. In this problem, $z = 2x_1 - x_2 + b$. Since the decision boundary is $z = 0$, we have.

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14 **[Intermediate]** • **Target: Dot Products, Cosine Similarity and Angular Distance**

Consider a simple linear regression model where the predicted value $\hat{y}$ is given by $\hat{y} = b_0 + b_1 x$. If we have two data points $(x_1, y_1) = (2, 5)$ and $(x_2, y_2) = (4, 9)$, and the Ordinary Least Squares (OLS) method yields the regression line $\hat{y} = 1 + 2x$, what is the predicted value $\hat{y}$ for $x = 3$?

The problem provides a simple linear regression equation $\hat{y} = b_0 + b_1 x$ and specific values for the intercept $b_0 = 1$ and the slope $b_1 = 2$. We are asked to find the predicted value $\hat{y}$ for a given value of x. We substitute $x = 3$ into the regression equation: $\hat{y} = 1 + 2(3)$ $\hat{y} = 1 + 6$ $\hat{y} = 7$ Therefore, the predicted value $\hat{y}$ for $x = 3$ is 7. The data points $(2, 5)$ and $(4, 9)$ are consistent with this regression line: for $x = 2$, $\hat{y} = 1 + 2(2) = 5$, and for.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: High-Dimensional Geometry and Projections** • **Cross-Lesson: 2.01 Neural Layer Matrices and Activations**

Consider a simple logistic regression model where the decision boundary is defined by the linear equation $w_1 x_1 + w_2 x_2 + b = 0$. If the weights are $w_1 = 2$ and $w_2 = 3$, and the bias is $b = -7$, what is the equation of the decision boundary?

The decision boundary in a logistic regression model is defined by the equation where the log-odds are zero. This corresponds to a linear combination of the input features and weights, plus the bias, being equal to zero. Given the weights $w_1 = 2$ and $w_2 = 3$, and the bias $b = -7$, the equation of the decision boundary is $w_1 x_1 + w_2 x_2 + b = 0$. Substituting the given values, we get $2x_1 + 3x_2 + (-7) = 0$, which simplifies to $2x_1 + 3x_2 - 7 = 0$. Final Answer: $2x_1 + 3x_2 - 7 = 0$

16 **[Challenging]** • **Target: Vectors & Embeddings in AI Practice Exercises** • **Cross-Lesson: 3.01 Gradient Descent and Optimization**

Consider a simple linear regression model where a dependent variable y is predicted from an independent variable x. The model is given by the equation $y = \beta_0 + \beta_1 x + \varepsilon$, where ε is the error term. Suppose we have estimated the regression line to be $\hat{y} = 3 + 2x$. If we observe an independent variable value of $x = 4$, what is the predicted value $\hat{y}$ for the dependent variable?

The problem asks for the predicted value $\hat{y}$ given an independent variable value $x = 4$ and a linear regression equation $\hat{y} = 3 + 2x$. We substitute the value of x into the equation: $\hat{y} = 3 + 2(4)$ $\hat{y} = 3 + 8$ $\hat{y} = 11$ Therefore, the predicted value for the dependent variable is 11. Common errors might involve incorrectly multiplying 2 by 4 or incorrectly adding 3 to 8.

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17 [Challenging] • Target: Vector Spaces and Linear Combinations • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

In a multiple linear regression model, the Sum of Squared Errors is $\text{SSE} = 20$ and the Total Sum of Squares is $\text{SST} = 100$. What is the coefficient of determination R^2?

The coefficient of determination measures the proportion of variance explained: $R^2 = 1 - \frac{\text{SSE}}{\text{SST}} = 1 - \frac{20}{100} = 1 - 0.20 = 0.80$. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

18 [Challenging] • Target: Vector Norms and Sparsity • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

A vector $\mathbf{v}$ is represented in a 2D space. If the components of the vector are $v_x = -3$ and $v_y = 4$, what is the Euclidean (L_2) norm of this vector?

The Euclidean (L_2) norm of a vector $\mathbf{v} = [v_x, v_y]^T$ is calculated as $||\mathbf{v}||_2 = \sqrt{v_x^2 + v_y^2}$. For the given vector $\mathbf{v}$ with components $v_x = -3$ and $v_y = 4$, the norm is: $||\mathbf{v}||_2 = \sqrt{(-3)^2 + (4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5$.

19 [Challenging] • Target: Dot Products, Cosine Similarity and Angular Distance • Cross-Lesson: 6.01 Linear Regression and OLS

Consider a 2D data point $\mathbf{v} = [3;\ -4]$. What is the squared Euclidean norm (or squared magnitude) of this vector?

The squared Euclidean norm of a vector $\mathbf{v} = [v_1; v_2]$ is given by $||\mathbf{v}||^2 = v_1^2 + v_2^2$. For the given vector $\mathbf{v} = [3;\ -4]$, the squared Euclidean norm is $(3)^2 + (-4)^2 = 9 + 16 = 25$.

20 [Challenging] • Target: High-Dimensional Geometry and Projections • Cross-Lesson: 7.01 Logistic Regression and Classification

A 2D loss surface is defined as $L(w_1, w_2) = 2w_1^2 + 3w_2^2$. Find the gradient vector $abla L$ at the point $(w_1, w_2) = (2,\ -1)$.

Compute partial derivatives: $\frac{\partial L}{\partial w_1} = 2(2)w_1 = 4(2) = 8$; $\frac{\partial L}{\partial w_2} = 2(3)w_2 = 6(-1) = -6$. Therefore, $abla L = [8,\ -6]^T$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Vectors & Embeddings in AI Practice Exercises

Compute the dot product of the embedding vector $\mathbf{u} = [2, 3]^T$ with itself.

a. 13 ✓

b. 15

c. 11

d. 10

The dot product with itself is the squared Euclidean norm: $\mathbf{u} \cdot \mathbf{u} = (2)^2 + (3)^2 = 4 + 9 = 13$.

22 [Foundational] • Target: Vectors & Embeddings in AI Practice Exercises

Compute the dot product of the embedding vector $\mathbf{u} = [3, 4]^T$ with itself.

a. 25 ✓

b. 29

c. 23

d. 14

The dot product with itself is the squared Euclidean norm: $\mathbf{u} \cdot \mathbf{u} = (3)^2 + (4)^2 = 9 + 16 = 25$.

23 [Foundational] • Target: Vectors & Embeddings in AI Practice Exercises

Compute the dot product of the embedding vector $\mathbf{u} = [4, 5]^T$ with itself.

a. 41 ✓

b. 47

c. 39

d. 18

The dot product with itself is the squared Euclidean norm: $\mathbf{u} \cdot \mathbf{u} = (4)^2 + (5)^2 = 16 + 25 = 41$.

24 [Foundational] • Target: Vectors & Embeddings in AI Practice Exercises

Compute the dot product of the embedding vector $\mathbf{u} = [5, 6]^T$ with itself.

a. 61 ✓

b. 69

c. 59

d. 22

The dot product with itself is the squared Euclidean norm: $\mathbf{u} \cdot \mathbf{u} = (5)^2 + (6)^2 = 25 + 36 = 61$.

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25 [Foundational] • Target: Vectors & Embeddings in AI Practice Exercises

Compute the dot product of the embedding vector $\mathbf{u} = [6, 7]^T$ with itself.

a. **85** ✓

b. 95

c. 83

d. 26

The dot product with itself is the squared Euclidean norm: $\mathbf{u} \cdot \mathbf{u} = (6)^2 + (7)^2 = 36 + 49 = 85$.

26 [Foundational] • Target: Vectors & Embeddings in AI Practice Exercises

Compute the dot product of the embedding vector $\mathbf{u} = [7, 8]^T$ with itself.

a. 111

b. 125

c. **113** ✓

d. 30

The dot product with itself is the squared Euclidean norm: $\mathbf{u} \cdot \mathbf{u} = (7)^2 + (8)^2 = 49 + 64 = 113$.

27 [Foundational] • Target: Vectors & Embeddings in AI Practice Exercises

Compute the Manhattan (L_1) norm and Euclidean (L_2) norm of the word vector $\mathbf{w} = [2, -3]^T$.

a. **$\|\mathbf{w}\|_1 = 5, \; \|\mathbf{w}\|_2 = \sqrt{13}$** ✓

b. $\|\mathbf{w}\|_1 = -1, \; \|\mathbf{w}\|_2 = 13$

c. $\|\mathbf{w}\|_1 = 5, \; \|\mathbf{w}\|_2 = 5$

d. $\|\mathbf{w}\|_1 = \sqrt{13}, \; \|\mathbf{w}\|_2 = 5$

L_1 norm is the sum of absolute values: $|2| + |-3| = 5$. L_2 norm is the square root of sum of squares: $\sqrt{(2)^2 + (-3)^2} = \sqrt{4 + 9} = \sqrt{13}$.

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28 [Foundational] • Target: Vectors & Embeddings in AI Practice Exercises

Compute the Manhattan (\(L_1 \)) norm and Euclidean (\(L_2 \)) norm of the word vector \(\mathbf{w} = [4, -6]^T \).

a. \(\|\mathbf{w}\|_1 = -2, \; \|\mathbf{w}\|_2 = 52 \)

b. **\(\|\mathbf{w}\|_1 = 10, \; \|\mathbf{w}\|_2 = \sqrt{52} \)** ✓

c. \(\|\mathbf{w}\|_1 = 10, \; \|\mathbf{w}\|_2 = 10 \)

d. \(\|\mathbf{w}\|_1 = \sqrt{52}, \; \|\mathbf{w}\|_2 = 10 \)

\(L_1 \) norm is the sum of absolute values: \(|4| + |-6| = 10 \). \(L_2 \) norm is the square root of sum of squares: \(\sqrt{(4)^2 + (-6)^2} = \sqrt{16 + 36} = \sqrt{52} \).

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Vector Spaces and Linear Combinations

Given word vectors $\mathbf{u} = [2, -1, 3]^T$ and $\mathbf{v} = [1, 4, -2]^T$, compute the linear combination $3\mathbf{u} - 2\mathbf{v}$.

a. $[4, -11, 13]^T$ ✓

b. $[4, 5, 5]^T$

c. $[8, -11, 5]^T$

d. $[7, 5, 13]^T$

$3[2, -1, 3] = [6, -3, 9]$. $2[1, 4, -2] = [2, 8, -4]$. Difference: $[6-2, -3-8, 9-(-4)] = [4, -11, 13]^T$.

30 [Intermediate] • Target: Vector Spaces and Linear Combinations

Two embedding vectors $\mathbf{a} = [k, 2, -1]^T$ and $\mathbf{b} = [3, -4, 2]^T$ are orthogonal in semantic space. Find the value of k.

a. $k = \frac{10}{3}$ ✓

b. $k = 2$

c. $k = -\frac{10}{3}$

d. $k = 5$

Orthogonal $\Rightarrow \mathbf{a} \cdot \mathbf{b} = 3k - 8 - 2 = 3k - 10 = 0 \Rightarrow k = 10/3$.

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31 [Intermediate] • Target: Vector Spaces and Linear Combinations

Determine whether the three feature vectors $\mathbf{v}_1 = [1, 0]^T$, $\mathbf{v}_2 = [0, 1]^T$, and $\mathbf{v}_3 = [3, -2]^T$ are linearly independent in $\mathbf{R}^2$.

a. Linearly dependent (any set of > 2 vectors in $\mathbf{R}^2$ is dependent) ✓

b. Linearly independent

c. Orthogonal set

d. Cannot be determined without a matrix inversion

Since the dimension of the space is 2, any collection of 3 vectors is linearly dependent: $\mathbf{v}_3 = 3\mathbf{v}_1 - 2\mathbf{v}_2$.

32 [Intermediate] • Target: Vector Norms and Sparsity

Calculate the Manhattan (L_1) norm and Euclidean (L_2) norm of the embedding error vector $\mathbf{e} = [3, -4, 12]^T$.

a. $||\mathbf{e}||_1 = 19, \quad ||\mathbf{e}||_2 = 13$ ✓

b. $||\mathbf{e}||_1 = 13, \quad ||\mathbf{e}||_2 = 19$

c. $||\mathbf{e}||_1 = 11, \quad ||\mathbf{e}||_2 = 13$

d. $||\mathbf{e}||_1 = 19, \quad ||\mathbf{e}||_2 = 169$

$L_1 = |3| + |-4| + |12| = 19$. $L_2 = \sqrt{3^2 + (-4)^2 + 12^2} = \sqrt{9 + 16 + 144} = \sqrt{169} = 13$.

33 [Intermediate] • Target: Vector Norms and Sparsity

Find the Chebyshev (L_∞) norm of the parameter vector $\mathbf{w} = [-7, 2, 5, -9, 4]^T$.

a. 9 ✓

b. 7

c. 27

d. 5

$||\mathbf{w}||_\infty = \max(|-7|, |2|, |5|, |-9|, |4|) = 9$.

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34 [Intermediate] • Target: Vector Norms and Sparsity

In Lasso regression, why does adding an L_1 penalty to the loss function induce exact feature sparsity (weights driven to 0)?

a. The diamond-shaped L_1 contour has sharp corners on the coordinate axes ✓

b. The L_1 norm is continuously differentiable everywhere

c. The L_1 penalty penalizes large weights much more heavily than L_2

d. The L_1 norm computes the determinant of the covariance matrix

The level sets of the L_1 ball are non-smooth polytopes with vertices on the axes, where elliptical loss contours naturally contact first.

35 [Intermediate] • Target: Dot Products, Cosine Similarity and Angular Distance

Two query embeddings are $\mathbf{u} = [1, 1, 0]^T$ and $\mathbf{v} = [0, 2, 2]^T$. Compute the cosine similarity $\cos\theta$ between them.

a. 0.500 ✓

b. 0.707

c. 0.250

d. 1.000

$\mathbf{u} \cdot \mathbf{v} = 2$. $||\mathbf{u}|| = \sqrt{2}$, $||\mathbf{v}|| = \sqrt{8} = 2\sqrt{2}$. $\cos\theta = \frac{2}{\sqrt{2}(2\sqrt{2})} = \frac{2}{4} = 0.5$.

36 [Intermediate] • Target: Dot Products, Cosine Similarity and Angular Distance

If two unit-normalized embeddings have cosine similarity $\cos\theta = -1.0$, what is their semantic relationship?

a. Diametrically opposite in direction (180° apart) ✓

b. Identical in meaning

c. Orthogonal / unrelated

d. Collinear in the exact same direction

$\cos(180^\circ) = -1.0$: the vectors point in exact opposite directions.

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37 [Intermediate] • Target: Dot Products, Cosine Similarity and Angular Distance

Compute the cosine distance $D_{\cos}(\mathbf{u}, \mathbf{v}) = 1 - \cos\theta$ for two orthogonal feature vectors.

a. 1.0 ✓

b. 0.0

c. 2.0

d. -1.0

For orthogonal vectors, $\cos\theta = 0$. Thus $D_{\cos} = 1 - 0 = 1.0$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: High-Dimensional Geometry and Projections • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

Find the scalar projection of vector $\mathbf{a} = [4, 2]^T$ onto the direction of unit vector $\mathbf{u} = [0.6, 0.8]^T$.

a. 4.0 ✓

b. 5.0

c. 2.4

d. 3.2

Scalar projection $= \mathbf{a} \cdot \mathbf{u} = 4(0.6) + 2(0.8) = 2.4 + 1.6 = 4.0$.

39 [Challenging] • Target: High-Dimensional Geometry and Projections • Cross-Lesson: 6.01 Linear Regression and OLS

In high-dimensional embedding spaces ($D = 1024$), why are two randomly selected vectors almost certainly nearly orthogonal?

a. **The surface area of a high-dimensional hypersphere concentrates near the equator relative to any pole** ✓

b. All high-dimensional vectors have zero norm

c. The dot product of random Gaussian variables always equals 1

d. The cosine function approaches infinity as dimension increases

By concentration of measure on the sphere $\mathbf{S}^{D-1}$, almost all mass lies in a thin band around the equator, so $\cos\theta \approx 0$ with high probability.

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40 [Challenging] • Target: High-Dimensional Geometry and Projections • Cross-Lesson: 7.01 Logistic Regression and Classification

Normalize the feature vector $\mathbf{x} = [3, 0, 4]^T$ to have unit Euclidean norm.

a. $[0.6, 0, 0.8]^T$ ✓

b. $[0.75, 0, 1]^T$

c. $[1, 0, 1]^T$

d. $[0.3, 0, 0.4]^T$

$||\mathbf{x}|| = \sqrt{3^2 + 0^2 + 4^2} = 5$. Unit vector $= \frac{1}{5}[3, 0, 4]^T = [0.6, 0, 0.8]^T$.

41 [Challenging] • Target: Vectors & Embeddings in AI Practice Exercises • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

Two document embeddings $\mathbf{a} = [6, 12]^T$ and $\mathbf{b} = [12, 6]^T$ are compared. If $||\mathbf{a}|| = ||\mathbf{b}|| = \sqrt{180}$, calculate their exact cosine similarity.

a. 1.000

b. 0.640

c. 0.800 ✓

d. 0.500

Dot product $\mathbf{a} \cdot \mathbf{b} = (6)(12) + (12)(6) = 144$. Product of norms $||\mathbf{a}||||\mathbf{b}|| = 180$. Cosine similarity = $\frac{144}{180} = 0.800$.

42 [Challenging] • Target: Vector Spaces and Linear Combinations • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

Two document embeddings $\mathbf{a} = [2, 4]^T$ and $\mathbf{b} = [4, 2]^T$ are compared. If $||\mathbf{a}|| = ||\mathbf{b}|| = \sqrt{20}$, calculate their exact cosine similarity.

a. 1.000

b. 0.640

c. 0.800 ✓

d. 0.500

Dot product $\mathbf{a} \cdot \mathbf{b} = (2)(4) + (4)(2) = 16$. Product of norms $||\mathbf{a}||||\mathbf{b}|| = 20$. Cosine similarity = $\frac{16}{20} = 0.800$.

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43 [Challenging] • Target: Vector Norms and Sparsity • Cross-Lesson: 3.01 Gradient Descent and Optimization

Two document embeddings $\mathbf{a} = [3, 6]^T$ and $\mathbf{b} = [6, 3]^T$ are compared. If $||\mathbf{a}|| = ||\mathbf{b}|| = \sqrt{45}$, calculate their exact cosine similarity.

a. 1.000

b. 0.640

c. 0.800 ✓

d. 0.500

Dot product $\mathbf{a} \cdot \mathbf{b} = (3)(6) + (6)(3) = 36$. Product of norms $||\mathbf{a}||||\mathbf{b}|| = 45$. Cosine similarity = $\frac{36}{45} = 0.800$.

44 [Challenging] • Target: Dot Products, Cosine Similarity and Angular Distance • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

Two document embeddings $\mathbf{a} = [4, 8]^T$ and $\mathbf{b} = [8, 4]^T$ are compared. If $||\mathbf{a}|| = ||\mathbf{b}|| = \sqrt{80}$, calculate their exact cosine similarity.

a. 1.000

b. 0.640

c. 0.800 ✓

d. 0.500

Dot product $\mathbf{a} \cdot \mathbf{b} = (4)(8) + (8)(4) = 64$. Product of norms $||\mathbf{a}||||\mathbf{b}|| = 80$. Cosine similarity = $\frac{64}{80} = 0.800$.

45 [Challenging] • Target: High-Dimensional Geometry and Projections • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

Two document embeddings $\mathbf{a} = [1, 2]^T$ and $\mathbf{b} = [2, 1]^T$ are compared. If $||\mathbf{a}|| = ||\mathbf{b}|| = \sqrt{5}$, calculate their exact cosine similarity.

a. 1.000

b. 0.640

c. 0.800 ✓

d. 0.500

Dot product $\mathbf{a} \cdot \mathbf{b} = (1)(2) + (2)(1) = 4$. Product of norms $||\mathbf{a}||||\mathbf{b}|| = 5$. Cosine similarity = $\frac{4}{5} = 0.800$.

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